

Jacobians in Other Frames

EGR 445/545

GVSU

plan for today

- inverse kinematics review example
- hints on LA 13
- debrief LA 11

Inverse Kin. Review Example

`review_inverse_kin_example.ipynb`

LA 13 Hint

Three link planar Jacobian:

if we use v_{tipz} as the third DOF, what will the bottom row of the Jacobian be?

Debrief LA 11

- why does this *almost* work?
 - what's the risk?

$${}^{w}_{tip}\mathbf{T}_{sym} = \begin{bmatrix} -\sin(\theta_4)\cos(\theta_5) & \sin(\theta_4)\sin(\theta_5) & \cos(\theta_4) & 9\cos(\theta_4) \\ \cos(\theta_4)\cos(\theta_5) & -\sin(\theta_5)\cos(\theta_4) & \sin(\theta_4) & 9\sin(\theta_4) \\ \sin(\theta_5) & \cos(\theta_5) & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

```
c_th4 = T_wtip_num[0,0]  
s_th4 = T_wtip_num[1,0]  
th4 = np.arctan2(s_th4,c_th4)*rtd
```